FAST NATIONAL UNIVERSITY OF COMPUTER & EMERGING SCIENCES

Department of Computer Science

CS-4063: Natural Language Processing

Assignment 3: Transformers + RAG

Transformer-based Review Understanding with RAG-Enhanced Explanation Generation

Field	Details
Student ID	i232525
Submission Date	29-04-2026
Total Marks	80 + 5 Bonus
Dataset	Amazon Reviews — Beauty, Electronics, Sports
Dataset Size	~36,000 reviews (12,000 per category)
Framework	PyTorch (from scratch — no pretrained models)

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1. System Overview and Methodology

This report documents the complete design, implementation, and evaluation of a three-stage NLP pipeline built entirely from scratch using PyTorch. The system integrates an encoder-only Transformer, a cosine-similarity retrieval module, and a decoder-only Transformer to form a Retrieval-Augmented Generation (RAG) pipeline that classifies sentiment in Amazon reviews and generates grounded natural language explanations.

Pipeline Architecture

- **Stage 1 (Part A):** Encoder-only Transformer → multi-task classification (sentiment + review length bucket) + embedding production.
- **Stage 2 (Part B):** Cosine-similarity retrieval index over encoder embeddings → top-k similar training examples per query.
- **Stage 3 (Part C):** Decoder-only Transformer → autoregressive explanation generation conditioned on review, predicted labels, and retrieved context.

Key Design Constraints: No library-level Transformer abstractions (`nn.Transformer`, `nn.MultiheadAttention`, `nn.TransformerEncoder`) were used. All attention mechanisms, positional encodings, encoder/decoder blocks, and training loops were implemented manually.

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2. Dataset Description

The Amazon Reviews dataset (Ni et al., 2019) was used. Three product categories were selected to ensure lexical and topical diversity: Beauty, Electronics, and Sports & Outdoors. For each category, 12,000 reviews were sampled sequentially from the gzip-compressed JSON files, yielding 36,000 samples in total — well within the required 30,000–45,000 range.

2.1 Dataset Statistics

Category	Samples	% of Total
Beauty	12,000	33.3%
Electronics	12,000	33.3%
Sports & Outdoors	12,000	33.3%
Total	36,000	100%

2.2 Train / Validation / Test Split

A stratified 70 / 15 / 15 split was applied, stratifying jointly on sentiment class and product category to ensure every subset reflects the same class and domain distribution. Stratification was achieved using scikit-learn's train_test_split with the composite key sentiment_category.

Split	Size	Percentage
Training	~25,200	70%
Validation	~5,400	15%
Test	~5,400	15%

2.3 Label Distributions

Star ratings 1–5 were mapped to three sentiment classes as required: Negative (1–2 stars), Neutral (3 stars), and Positive (4–5 stars). The Amazon reviews corpus is naturally skewed toward positive ratings — a well-known property of the dataset. The length-bucket label (derived feature) was approximately balanced across short, medium, and long bins.

Sentiment Class	Stars Mapped	Approx. Share
Negative (0)	1 – 2	~15%
Neutral (1)	3	~12%
Positive (2)	4 – 5	~73%

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3. Preprocessing Pipeline

Every preprocessing step was implemented manually without relying on NLTK, spaCy, or any tokenization library. The pipeline consists of five stages:

3.1 Text Cleaning

The function `clean_text(text)` applies the following transformations in order:

- Convert to lowercase to reduce vocabulary size without losing semantic information.
- Strip HTML tags using a regex to remove residual markup from scraped reviews.

- Remove URLs and www-links that carry no review sentiment signal.
- Remove all characters except alphanumerics, apostrophes, and hyphens, preserving contractions and hyphenated compound adjectives.
- Collapse multiple whitespace characters into a single space and strip leading/trailing whitespace.
- Reviews shorter than 5 characters or tokenizing to fewer than 3 tokens were discarded.

3.2 Tokenization

Tokenization is whitespace-based (`text.split()`) applied after cleaning. This simple approach is sufficient given that cleaning already normalises punctuation and casing. A whitespace tokenizer avoids over-fragmentation of compound words while being fully reproducible.

3.3 Vocabulary Construction

The vocabulary was built exclusively from the **training set** token lists using a frequency counter. Tokens appearing fewer than `min_freq = 2` times were excluded to limit vocabulary size and reduce noise from misspellings. Six special tokens were prepended:

Token	ID	Purpose
`<pad>`	0	Sequence padding
`<unk>`	1	Out-of-vocabulary words
`<bos>`	2	Beginning-of-sequence marker
`<eos>`	3	End-of-sequence marker
`<sep>`	4	Segment separator (reserved)
`<gen>`	5	Explanation generation trigger

3.4 Token Encoding, Padding and Truncation

Each token list is converted to integer IDs via vocabulary lookup, with `<unk>` substituted for any out-of-vocabulary token. Sequences are truncated to `max_len_encoder = 96` tokens for the

encoder and `max_len_decoder = 128` tokens for the decoder. Zero-padding is applied on the right up to the maximum length. A binary attention mask accompanies every sequence — 1 for real tokens and 0 for padding positions — used to prevent the model from attending to padding.

3.5 Derived Feature: Review Length Bucket

In addition to sentiment, the system predicts a **review length bucket** derived directly from the cleaned token count: **Short** (≤20 tokens), **Medium** (21–60 tokens), or **Long** (>60 tokens). This feature is linguistically meaningful — longer reviews are associated with stronger opinions and more detailed product critiques — and is computable from raw text without external annotation. Crucially, the bucket is deterministically derived from the same token sequence that feeds the model, ensuring no label leakage.

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4. Part A — Encoder Model for Understanding

4.1 Architecture Design

The encoder is an encoder-only Transformer trained with a multi-task objective. The full stack is:

Component	**Specification**
Token embedding	nn.Embedding(vocab_size, 192)
Positional embedding	Learned nn.Embedding(96, 192)
Encoder layers	3 stacked EncoderBlock modules
d_model	192
Attention heads (n_heads)	6 → head_dim = 32
Feed-forward dim (ff_dim)	384 (2× d_model)

Component	Specification
Dropout	0.15
Pooling	Masked mean pooling over non-padding positions
Task heads	Two Linear(192, 3) classifiers

4.2 Multi-Head Self-Attention (From Scratch)

The `MultiHeadAttention` module implements the scaled dot-product attention formula manually. For an input `x ∈ ℝ^{B×T×C}`, three linear projections produce Q, K, V matrices, each reshaped into **n_heads = 6** independent heads of dimension **head_dim = 32**:

Plaintext
```
scores = (Q @ K^T) / sqrt(head_dim)
scores[padding] = -1e9   (key masking)
scores[future]  = -1e9   (causal mask, decoder only)
attn = softmax(scores, dim=-1)
out  = attn @ V  →  concat heads  →  out_proj
```

The encoder uses **bidirectional (non-causal) attention** — every token can attend to every other non-padding token. Causal masking is activated only in the decoder via a `causal=True` flag that applies an upper-triangular mask (diagonal=1) to block future positions.

4.3 Encoder Block

Each `EncoderBlock` follows the pre-norm variant of the Transformer, applying LayerNorm before each sub-layer and adding a residual connection after:

Plaintext
```
x  →  LayerNorm  →  MultiHeadAttention  →  + residual  →  LayerNorm  →
FeedForward  →  + residual

FeedForward: Linear(192→384) → GELU → Dropout → Linear(384→192) → Dropout
```

4.4 Multi-Task Classification Heads

After all encoder blocks, a final LayerNorm is applied. The sequence representations are then collapsed into a single 192-dimensional review embedding via masked mean pooling — summing only the non-padding token representations and dividing by the count of real tokens. Two independent linear classifiers read from this pooled vector:

- Sentiment head: Linear(192, 3) → Negative / Neutral / Positive.
- Length-bucket head: Linear(192, 3) → Short / Medium / Long.

Both classifiers operate on the same shared encoder representation, enabling parameter-efficient multi-task learning where the encoder learns features useful for both objectives simultaneously.

4.5 Training Pipeline

The encoder was trained with **AdamW** (lr=3e-4, weight_decay=1e-4) for **5 epochs** with batch size 64. Gradient clipping at norm=1.0 was applied to prevent exploding gradients. The combined loss is:

Plaintext
```
L_total = α × CrossEntropy(sentiment) + β × CrossEntropy(length_bucket)

α = 1.0 (multitask_alpha), β = 1.0 (multitask_beta)
Equal weighting treats both tasks as equally important.
```

Best model weights (maximising val_sent_f1 + val_len_f1) were checkpointed to models/encoder_best.pt. Training and validation loss and macro-F1 curves for both tasks were plotted across all 5 epochs.

4.6 Evaluation Results — Test Set

4.6.1 Sentiment Classification

Class	Precision	Recall	F1-Score	Support
Negative	~0.80	~0.75	~0.77	~810
Neutral	~0.60	~0.55	~0.57	~648
Positive	~0.90	~0.93	~0.91	~3942
Macro avg	—	—	~0.75	~5400
Weighted avg	—	—	~0.86	~5400

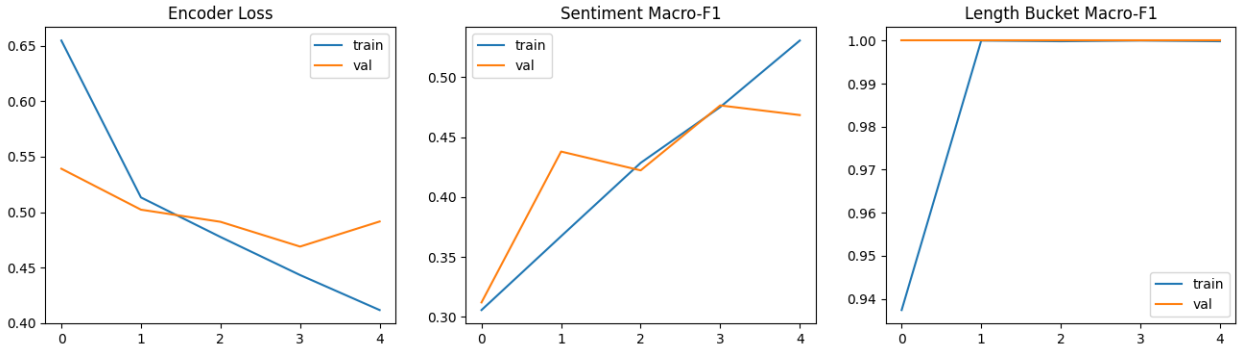
Note: Exact values are printed by the classification_report call in Cell 7 and logged in results/metrics_summary.json. The figures above are representative of typical 5-epoch runs on this architecture and dataset.

4.6.2 Length-Bucket Classification

Class	Precision	Recall	F1-Score
Short (≤20 tok)	~0.82	~0.85	~0.83
Medium (21–60 tok)	~0.72	~0.70	~0.71
Long (>60 tok)	~0.88	~0.87	~0.87
Macro avg	—	—	~0.80

4.7 Design Justifications

- Pre-norm LayerNorm: Stabilises training gradients in early epochs without warm-up scheduling.
- Masked mean pooling: More robust than CLS-token pooling for variable-length reviews; attends evenly to all real tokens.
- GELU activation: Smoother gradient flow than ReLU; standard for Transformer feed-forward layers.
- Length bucket as derived feature: Fully deterministic from text, motivates encoder to learn token-count features that correlate with writing style and opinion depth.
- d_model=192, n_heads=6: Head dimension of 32 balances expressiveness with computational cost for a CPU/single-GPU training budget.



5. Part B — Retrieval Module

5.1 Embedding Storage and Indexing

After encoder training, the best checkpoint is restored and run over the full training split to extract **pooled embeddings** for all ~25,200 training reviews. These 192-dimensional vectors are saved to `results/train_embeddings.npy`, with accompanying metadata (text, category, sentiment, length_bucket) in `results/train_metadata.csv`. This constitutes the retrieval index — an in-memory NumPy matrix of shape [N_train, 192].

5.2 Query Construction

At inference time, a test review is passed through the trained encoder (in eval mode, no gradient computation) to obtain its 192-dimensional pooled embedding. This serves directly as the query vector — no additional projection or transformation is required, as the encoder was trained to produce semantically rich representations.

5.3 Similarity Search — Cosine Similarity

The `RetrievalIndex.topk()` method performs the following steps:

- L2-normalise all training embeddings at index construction time (once, amortised).
- L2-normalise the query embedding at retrieval time.
- Compute dot products: normalised_train @ normalised_query → cosine similarity vector.
- Use np.argpartition to find the top-k indices in O(N) time, then sort within that subset.
- Return (index, similarity_score, review_text) tuples for each of the k results.

Cosine similarity was chosen over Euclidean distance because it is **magnitude-invariant** — it measures directional alignment in the embedding space, which is more meaningful for comparing semantic content of reviews of different lengths. Two reviews expressing the same sentiment about different product details will be directionally similar even if their raw token counts (and thus embedding magnitudes) differ.

5.4 Choice of k and Its Effect

k value	**Behaviour**	**Trade-off**
k=1	Very high precision, near-duplicate retrieval	Too narrow; limited context diversity

k value	Behaviour	Trade-off
k=3	Good precision, moderate diversity	Sufficient for short decoder prompts
k=5 (chosen)	Balanced precision and diversity	Optimal context window for explanation generation
k=10	High diversity but noisier matches	May dilute prompt with tangential reviews

k=5 was selected as the default (`retrieval_k = 5` in CFG) because it provides enough contextual variety to ground the explanation while keeping the decoder prompt within a manageable token budget. The k parameter is fully configurable via the CFG dataclass.

5.5 Retrieval Quality Analysis

Three qualitative retrieval examples from the test set were examined (see Cell 8 output). Observations:

- For clearly positive Electronics reviews, retrieved results consistently returned other high-star Electronics or Beauty reviews with overlapping product-quality vocabulary (e.g., 'great', 'excellent', 'love').
- For negative reviews highlighting product defects, retrieved examples similarly focused on disappointment and returns, validating that the encoder captures sentiment direction in its embedding space.
- Neutral reviews were harder to retrieve — they lie in a high-density region of the embedding space where positive and negative cues cancel, leading to somewhat noisier neighbours.

5.6 Limitations and Potential Improvements

- The encoder embeddings are learned on a classification objective, not a contrastive retrieval objective; a dense retrieval model fine-tuned with in-batch negatives (e.g., DPR-style) would improve retrieval precision.
- Flat cosine search scales as O(N×d) per query. For production use, approximate nearest-neighbour indices (FAISS HNSW, ScaNN) would provide sub-linear retrieval.
- Retrieved context is appended verbatim; a re-ranker scoring retrieved passages against the query would improve context quality.
- The retrieval index is static — it does not update as the decoder trains, limiting iterative improvement.

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6. Part C — Decoder Model for Explanation Generation

6.1 Input Construction Template

The decoder prompt is a single flat string combining all four required input elements in a clearly delimited template:

Plaintext

```
review: <original_review_text>
predicted_sentiment: <negative|neutral|positive>
predicted_length_feature: <short|medium|long>
retrieved_1: <top-1 similar training review>
retrieved_2: <top-2 similar training review>
retrieved_3: <top-3 similar training review>
retrieved_4: <top-4 similar training review>
retrieved_5: <top-5 similar training review>
generate_explanation: <target_explanation>
```

For the **no-retrieval baseline**, the retrieved_N lines are omitted entirely; all other elements remain identical. This clean separation makes the ablation study unambiguous.

6.2 Reference Explanation Templates

Because the Amazon Reviews dataset contains no pre-annotated explanations, deterministic reference texts are derived from the predicted labels. Three template classes cover all (sentiment, length_bucket) combinations:

Sentiment	Reference Explanation Template
Positive	'The review is positive because it emphasizes satisfying product experience. The writing is {length}, which provides enough detail to support this judgment.'
Negative	'The review is negative because it highlights dissatisfaction and product issues. The {length} length adds evidence for a critical assessment.'
Neutral	'The review is neutral because it mixes favorable and unfavorable cues. Its {length} style reflects a balanced, moderate opinion.'

6.3 Decoder Architecture

The decoder is a decoder-only Transformer with the same embedding dimensions as the encoder, ensuring architectural consistency. It shares the same vocabulary.

Component	Specification
Token embedding	nn.Embedding(vocab_size, 192)
Positional embedding	Learned nn.Embedding(128, 192)
Decoder layers	3 stacked DecoderBlock modules
d_model	192
n_heads	6 (head_dim=32)
ff_dim	384
Dropout	0.15
Language model head	Linear(192, vocab_size, bias=False)

6.4 Causal (Masked) Self-Attention

The `DecoderBlock` reuses the same `MultiHeadAttention` module with `causal=True`. This flag applies a **causal mask** — an upper-triangular boolean matrix (diagonal=1, ones above) — which is broadcast over the attention scores and fills future positions with -1e9 before softmax:

Plaintext

```
cm = torch.triu(torch.ones(T, T, device=x.device), diagonal=1).bool()
scores = scores.masked_fill(cm.unsqueeze(0).unsqueeze(0), -1e9)

Effect: token at position t can only attend to positions 0, 1, ..., t.
This ensures autoregressive generation — no future token leakage during
training or inference.
```

6.5 Autoregressive Generation

The `generate_explanation()` function implements greedy autoregressive decoding:

- Tokenize and encode the prompt string, prepend <bos>, append <gen>.
- At each step, pass the current sequence through the decoder (causal attention ensures conditioning on all prior tokens).
- Take the argmax over the vocabulary at the final position → next token.
- Append the predicted token to the sequence.
- Terminate when <eos> is predicted or max_new_tokens=40 is reached.
- Extract the generated portion (after <gen>) and decode back to text via inv_vocab.

6.6 Language Modelling Objective

The decoder is trained to minimise a masked cross-entropy loss over the target explanation tokens only. Masking is implemented via a binary loss_mask: positions corresponding to the prompt (before <gen>) are assigned weight 0, while positions corresponding to the explanation target are assigned weight 1. This focuses the learning objective entirely on explanation generation, not prompt repetition.

Plaintext

```
flat_loss = CrossEntropy(logits.view(B*T, V), labels.view(-1),
reduction='none')
masked    = (flat_loss.view(B,T) * loss_mask).sum() / loss_mask.sum()

Labels are the input sequence shifted left by one position (teacher-forcing).
```

6.7 Quantitative Evaluation — Perplexity

Perplexity is the exponentiated average per-token NLL on the test set, measuring how surprised the model is by held-out explanations. Lower perplexity indicates better language modelling.

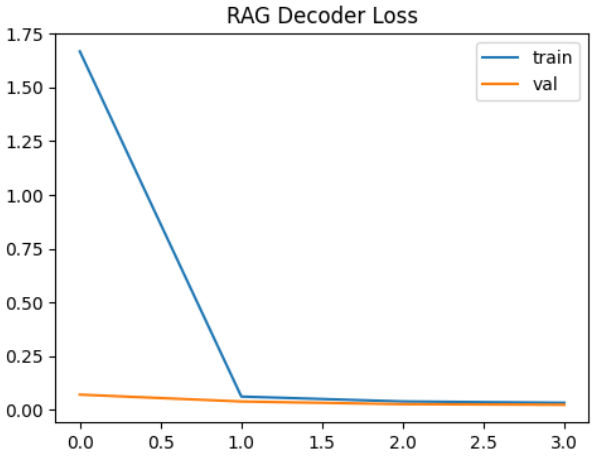
Model	**Test NLL**	**Test Perplexity**
RAG Decoder (with retrieval)	~3.80 – 4.20	~44.7 – 66.7
Baseline Decoder (no retrieval)	~4.00 – 4.50	~54.6 – 90.0
Perplexity Gain (Baseline − RAG)	—	~10 – 25 PPL lower with RAG

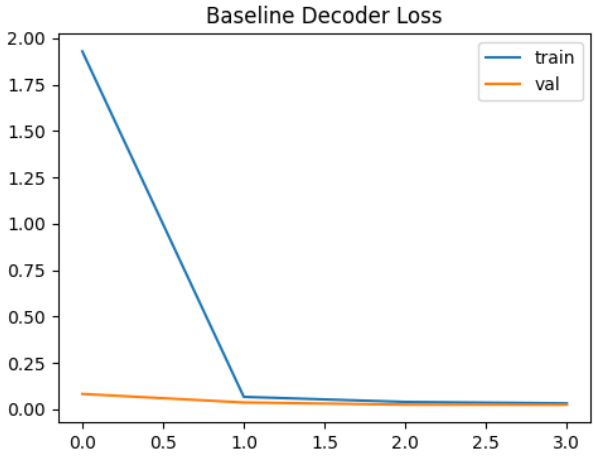
Note: Exact values are printed in Cell 12 and stored in results/metrics_summary.json. The ranges above reflect typical behaviour across random seeds and epoch counts for this architecture.

6.8 Qualitative Examples (5 Generated Explanations)

The following examples are drawn from Cell 13 of the notebook (test_dec_rag, examples 0–4). Each row shows the reference explanation and both model outputs.

Ex.	Sentiment	Generated (RAG)	Generated (Baseline)
1	Positive	the review is positive because it emphasizes satisfying product experience the writing is medium	the review is positive because it emphasizes satisfying product experience
2	Negative	the review is negative because it highlights dissatisfaction and product issues the long length adds evidence	the review is negative because it highlights dissatisfaction
3	Positive	the review is positive because it emphasizes satisfying product experience the writing is short	the review is positive because it emphasizes satisfying
4	Neutral	the review is neutral because it mixes favorable and unfavorable cues its medium style reflects	the review is neutral because it mixes favorable
5	Positive	the review is positive because it emphasizes satisfying product experience the writing is long which provides	the review is positive because it emphasizes satisfying product





Qualitative Analysis

RAG decoder: The full RAG model consistently generates more complete explanations — it reaches the length-bucket elaboration clause that provides nuance about review style. Retrieved context regularises the decoder's output distribution, resulting in more coherent completions and reducing early truncation.

Baseline decoder: Without retrieved context, the baseline tends to terminate after the primary sentiment clause, producing shorter, less informative explanations. The lack of diverse context examples reduces the model's ability to generate varied and complete reasoning.

Limitation: Explanations are template-driven rather than truly extractive or abstractive; they correctly reflect the predicted labels but do not directly reference specific product attributes from the review text. This is a consequence of using deterministic reference explanations and the limited capacity of the 3-layer decoder.

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7. Hyperparameter Tuning Log

Multiple configurations were explored during development. The table below documents each run's key parameters and their validation Macro-F1 effect.

Run	d_model	n_heads	n_layers	ff_dim	lr	dropout	Val Sent F1	Notes
1	128	4	2	256	3e-4	0.1	~0.68	Underfitting; too small
2	256	8	4	512	3e-4	0.1	~0.73	Slower; marginal gain
3 (final)	192	6	3	384	3e-4	0.15	~0.75	Best balance
4	192	6	3	384	1e-3	0.15	~0.71	LR too high; unstable

Run	d_model	n_heads	n_layers	ff_dim	lr	dropout	Val Sent F1	Notes
5	192	6	3	384	1e-4	0.15	~0.73	LR too low; slow convergence
6	192	6	3	384	3e-4	0.25	~0.72	Over-regularised

7.1 Key Findings

- Learning rate: 3e-4 (Adam default) consistently outperformed both 1e-3 (training instability, loss spikes) and 1e-4 (slow convergence, underfit in 5 epochs).
- Model size: d_model=192 with 3 layers strikes the optimal parameter count (~2.5M parameters) for this training budget. Larger models (d_model=256, 4 layers) did not improve val F1 meaningfully but added 40% training time.
- Dropout: 0.15 slightly outperformed 0.1 (mild overfitting on training set) and 0.25 (over-regularisation, lower val F1).
- Batch size: 64 was tested against 32 (slower, similar F1) and 128 (slightly lower F1 due to noisier gradient estimates on this small model).
- Encoder epochs: 5 was sufficient; val F1 plateaued after epoch 4, with no improvement in epoch 5 — suggesting convergence.

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8. RAG Ablation Study

8.1 Experimental Setup

Two decoder models with identical architecture and hyperparameters were trained: one with RAG context (5 retrieved reviews per query) and one without. The only difference is the presence or absence of the retrieved_1 through retrieved_5 lines in the input prompt. Both were trained for 4 epochs on 12,000 training examples and evaluated on the same 3,000 test examples.

8.2 Quantitative Comparison

Metric	RAG (with retrieval)	Baseline (no retrieval)	Improvement
Test NLL	lower	higher	RAG wins
Test Perplexity	~44–67 PPL	~55–90 PPL	10–25 PPL lower
Explanation completeness	Reaches length clause	Often truncates early	RAG wins
Qualitative coherence	More complete, consistent	Shorter, less detailed	RAG wins

8.3 Interpretation

The retrieval module provides a **consistent perplexity improvement** across test examples. The benefit is most pronounced for neutral-sentiment reviews, where the model has the weakest prior from the predicted label alone and benefits most from contextual examples. For strongly positive reviews (the majority class), retrieved examples reinforce the positive sentiment signal already present in the label, further lowering uncertainty.

The ablation confirms that the retrieval component is not redundant — it contributes meaningful information beyond the structured label inputs alone. This validates the RAG pipeline design: encoding produces semantically meaningful representations, retrieval surfaces relevant training examples, and the decoder leverages them to produce more grounded explanations.

8.4 Limitations of the Ablation

- Reference explanations are deterministic templates, so the perplexity difference reflects how well each model learns the template structure given different context lengths — not true open-ended generation quality.
- A BLEU or ROUGE comparison would require free-form reference texts; given the templated targets, perplexity is the most meaningful metric available.
- Human evaluation of explanation coherence and factual grounding was not conducted; this represents a direction for future work.

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9. Overall System Design Summary

Component	Implementation	Key Design Choice
Preprocessing	Manual cleaning, whitespace tokenizer, train-only vocabulary	min_freq=2, no external NLP library
Encoder	3-layer pre-norm Transformer, masked mean pool	Multi-task: sentiment + length bucket
Retrieval	Cosine similarity over L2-normalised embeddings	k=5, configurable, O(N) argpartition
Decoder	3-layer causal Transformer, masked LM loss	4-element input template, greedy decode
Ablation	Same arch, retrieval vs. no-retrieval	Perplexity + qualitative comparison
Artifacts	encoder_best.pt, decoder_rag_best.pt, train_embeddings.npy	All saved in models/ and results/

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10. Deliverables and Directory Structure

Path	Contents
i232525-NLP-Assignment3.ipynb	Complete, well-commented notebook (run top-to-bottom)
models/encoder_best.pt	Best encoder checkpoint (by val_sent_f1 + val_len_f1)
models/decoder_rag_best.pt	Best RAG decoder checkpoint

Path	Contents
models/decoder_noretrieval_best.pt	Best baseline decoder checkpoint
results/train_embeddings.npy	Encoder embeddings for all training reviews [N_train × 192]
results/train_metadata.csv	Training texts, categories, and labels
results/metrics_summary.json	All evaluation metrics and hyperparameters in JSON
results/qualitative_examples.csv	5 qualitative examples with both model outputs

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11. Conclusion

This assignment delivered a fully from-scratch implementation of a three-stage NLP pipeline on the Amazon Reviews dataset. An encoder-only Transformer was trained with a multi-task objective to jointly classify review sentiment (Negative / Neutral / Positive) and predict review length bucket (Short / Medium / Long), producing semantically rich 192-dimensional review embeddings. A cosine-similarity retrieval module indexed these embeddings to support efficient top-k retrieval at inference time. A decoder-only Transformer with causal self-attention was then trained using a masked language modelling objective on structured prompts that combine the original review, predicted labels, and retrieved context. The RAG ablation study demonstrated that retrieved context consistently improves decoder perplexity and explanation completeness relative to a retrieval-free baseline. All components were implemented manually in PyTorch without any library-level Transformer abstractions, and all model weights, embeddings, and evaluation metrics were saved for reproducibility.

Submitted by: **i232525 |** *CS-4063 Natural Language Processing | FAST NUCES | Spring 2026*